



Epidemiology and Outcome of Diabetic Foot Ulcers in Nigeria: A Systematic Review and meta-analysis of 2 Decades: 2005 – 2024

Michael Adeyemi Olamoyegun¹ · Samson Adedeji Afolabi² · Kehinde Alare² · Oluwadamilola Dolapo Olamoyegun² · Taiwo Adeyemi² · Oluwarotimi Bolaji Olopade³

Received: 15 October 2025 / Accepted: 22 December 2025
© The Author(s) 2026

Abstract

Introduction Diabetic foot ulcers (DFUs) are a main factor leading to morbidity, hospitalizations, amputations, disability, and death of people with diabetes. However, the epidemiology and outcomes of DFUs in Nigeria are not well documented. This study (systematic review and meta-analysis) aimed to evaluate the prevalence, amputation rates, and clinical outcomes of DFU patients across various regions in Nigeria.

Methods The search was comprehensively conducted through electronic databases: African Journal Online (AJOL), PubMed, SCOPUS, and Google Scholar, for studies published in English between 2005 and July 15, 2024. Eligible studies reporting on the prevalence and outcomes of DFUs in Nigerian populations were included. Some of the search terms used are (“diabetes” OR “diabetic”) AND (“foot” OR “lower extremity” OR “limb” or “leg”) AND (“amputation” OR “amputations” OR “infection” OR “gangrene” OR “necrosis” OR “ulcer” OR “ulcers” OR “wound” OR “wounds”). Two investigators independently assessed the methodological quality using the Newcastle–Ottawa Scale, and key data were extracted using Microsoft Excel. Meta-regression and meta-analysis were performed using Stata version 16, employing a random effects model to estimate pooled prevalence and outcomes. Heterogeneity among studies was assessed using the I^2 statistic.

Result Eighteen full studies were included, encompassing 2,616 patients with DFU across the studies. The mean age was 56.43 years, with average diabetes duration of 10.62 years and ulcer duration of 46.65 days. The pooled prevalence of DFUs was 16.69% (95% CI 9.74–25.06), with the highest prevalence in the Southwest (20.38%, 95% CI 7.13–38.11) and the lowest in the Northeast (5.98%, 95% CI 4.86–7.20). DFUs predominantly affected the left foot (55.09%). The overall amputation prevalence was 33.27% (95% CI 23.09–44.29), with below-knee amputations being the most common (39.81%). The pooled mortality rate was 17.75% (95% CI 10.94–25.74), and 11.52% of patients left the hospital against medical advice. The average length of stay (on hospital admission) of the entire study population was 39.97 days.

Conclusion DFUs are a high health burden in Nigeria, with high rates of amputation and mortality that vary by region. These regional differences indicate an urgent need for targeted interventions. A multidisciplinary approach to DFU management is essential to reduce amputation rates and improve health outcomes.

Systematic Review Registration <https://www.crd.york.ac.uk/prospero>, identifier CRD42024573581.

Keywords Diabetic foot ulcers · Amputations · Prevalence · Outcome · Nigeria

✉ Michael Adeyemi Olamoyegun
maolamoyegun@lautech.edu.ng

¹ Department of Medicine, Endocrinology, Diabetes & Metabolism Unit, Ladoke Akintola University of Technology/LAUTECH Teaching Hospital, Ogbomoso, Oyo State, Nigeria

² Department of Medicine, Ladoke Akintola University of Technology, Ogbomoso, Oyo State, Nigeria

³ Department of Medicine, Lagos University Teaching Hospital, Idi-Araba, Lagos, Nigeria

1 Introduction

Diabetes mellitus (DM) is known as the most prevalent metabolic disorder, affecting 9.3% (589 million) of the global population in 2024, with projections indicating an increase to 12% (853 million) by 2050 [1]. Individuals with diabetes are at a heightened risk of experiencing a variety of acute and chronic complications, with lower extremity infections and amputations being among the most feared and debilitating [2]. Each year, around 18.6 million individuals with

diabetes worldwide experience the development of a foot ulcer [3], and between 25 and 34% of all individuals with either type 1 or type 2 diabetes are expected to encounter a foot ulcer during their lifetime [2]. Diabetic foot ulcers contribute significantly to the burden of global disability. It is estimated that nearly 20% of all diabetic foot ulcers (DFUs) ultimately result in amputations [4]. Lower-extremity complications related to diabetes are responsible for up to 80% of all amputations in that area and are a significant cause of hospitalization globally. In Africa, the prevalence of diabetic foot ulcers (DFU) has been seen to vary between 10% and 30%. The incidence of DFU is notably higher in rural areas compared to urban centers. This increased occurrence is attributed to the prevalent practice of walking barefoot in rural communities. Additionally, the limited financial resources in these regions hinder individuals with diabetes from obtaining suitable footwear, exacerbating the problem. Given the anticipated rise in diabetes prevalence as predicted by the International Diabetes Federation (IDF), an increase in diabetic foot ulcer cases is likely to follow. The prevalence of diabetic foot ulcers (DFU) among hospitalized individuals in Nigeria has been reported to be significantly high, ranging from 11 to 32%, with variations observed across different regions [5]. Furthermore, DFU has been identified as the leading cause of diabetes-related deaths in Nigeria, following hyperglycemic emergencies [5]. Consequently, diabetic foot ulceration presents a critical public health issue in Nigeria.

The results for hospitalized patients with DFU may include complete or partial wound healing after varying lengths of time, leaving the hospital against medical advice (LAMA), limb amputation, or death. In the United States, over 150,000 non-traumatic lower limb amputations are conducted annually among individuals with diabetes [6]. Globally, around 1.6 million amputations take place each year, with approximately 33% classified as major amputations [7, 8]. The likelihood of death is approximately 2.5 times greater in diabetic individuals with foot ulcers compared to those without them [2]. The probability of lower-limb amputation is reported to be 15 to 46 times higher among individuals with diabetes mellitus (DM) than in those without it. Mortality rates related to the emergence of a diabetic foot ulcer (DFU) are estimated to be 5% within the first year and over 40% within five years [9]. For individuals undergoing major amputations, such as above-knee amputations, the five-year mortality rate has been observed to range from 42% to 75% [3, 10–12]. Despite the growing concern surrounding DFUs, there is a notable lack of contemporary data regarding the actual impact of DFUs in Nigeria. Additionally, most of the existing studies on this topic were conducted at a single center, showed significant variability in study design and demographic populations, were primarily retrospective,

and originated from various geopolitical regions, with none being nationwide. To address this deficiency, we aimed to perform a current, nationwide, and thorough assessment of the epidemiology and outcomes of diabetic foot ulcers in Nigeria through a systematic review and meta-analysis of all pertinent studies. The findings gathered could inform the creation of strategies for preventing and treating diabetic foot ulcers, enhance treatment outcomes, improve patients' quality of life, and alleviate the economic impact.

2 Methodology

2.1 Study Area

Nigeria is one of the countries that make up Western Africa, with a total area of 910,770 square kilometers, it accommodates almost 250 different ethnic groups [13]. Its 36 states are divided into 6 geo-political zones, along with a capital territory, Abuja. Nigeria's population was estimated to be 233 million thus making it the sixth most populous country in the world representing 2.78% of the global population [14, 15].

2.2 Search Strategy

We searched English-language databases including PubMed, African Journal Online (AJOL), Scopus and Google Scholar for cross-sectional, prospective, and observational studies reporting the epidemiology, clinical characteristics, and outcomes of diabetic patients with foot ulcers in Nigeria. Grey articles were also searched manually. The search string was: ("diabetes" OR "diabetic") AND ("foot" OR "lower extremity" OR "limb" or "leg") AND ("amputation" OR "amputations" OR "infection" OR "gangrene" OR "necrosis" OR "ulcer" OR "ulcers" OR "wound" OR "wounds") AND ("patient" OR "patients") AND "epidemiology" OR "incidence" OR "prevalence" AND "Nigeria". All the selected databases were searched till 15th July 2024 and the articles included were reviewed to identify other possible relevant articles. Wagner classification was used to assess the severity of the ulcer. Major amputation was defined as amputation done above the level of the ankle. Studies done among individuals with diabetes but without foot complications were excluded.

2.3 Study Design

This systematic review was conducted in July 2024 following the Preferred Reporting Items for Systematic Review and Meta-Analyses (PRISMA) 2021 guidelines [15, 16]. The search was performed after the protocol was accepted

on the International Prospective Register of Systematic Reviews (PROSPERO) with the following identification number: CRD42024573581.

2.4 Research Question

The PICO framework was used for systematic reviews in line with the literature [17], as we aimed to assess data on the burden of diabetic foot ulcers and the outcomes of its management.

Population (P): Patients with diabetes who developed foot ulcers.

Intervention (I): Not applicable.

Comparators (C): Not applicable.

Outcomes (O): Prevalence and treatment outcomes.

Thus, our research questions were as follows:

1. What is the pooled prevalence of foot ulcers among patients with diabetes in Nigeria and its variations across geopolitical zones?
2. What are the diabetic foot ulcer treatment outcomes and their predictors in Nigeria?

The Briggs Institute Reviewer manual approach, which consists of five steps—defining the research question, locating relevant studies, selecting studies, organizing data, and compiling, summarizing, and reporting the findings—was used to conduct the systematic review and meta-analysis [18].

2.5 Eligibility Criteria

Studies were included in the review if they met the following criteria: (1) hospital-based research with information on prevalence, potential predisposing factors, and treatment outcomes of diabetic foot ulcers (2) studies in which the full text was published in the English language; (3) original research. We excluded studies that were (1) solely based on self-reported diagnosis of diabetic foot ulcer; (2) studies in which a clear diagnosis of diabetic foot ulcer was not made (3) inclusion of foot at risk as part of analysis (4) studies such as RCTs, quasi-RCTs, crossover trials, controlled before and after studies, studies of simulation models, (5) case series, reviews, commentaries, experts' opinion, conference proceedings, letters, or editorials.

2.6 Data Extraction

Two reviewers worked in tandem to search the literature and evaluate relevant studies, using an eligibility guideline to guarantee that the selection criteria were consistently followed. Information was retrieved on the study's location,

duration and publication year, design, study setting (rural or urban), sample size, diagnostic standards, and population mean age. For a methodical evaluation, the extracted data were arranged according to Nigeria's geopolitical zones and saved in Microsoft Excel file format.

2.7 Quality Assessment

The Newcastle–Ottawa Scale (NOS), which was used to grade the quality of the research, has the following evaluation criteria: nonresponse rate, representativeness of the cases, sufficient case definition, and identical ascertainment method. In terms of quality assessment, the NOS' overall score ranged from 0 stars (lowest quality) to 9 stars (highest quality). If a study earned seven or more stars, it was frequently regarded as high-quality, moderate risk of bias (4–6) and <4 are considered to be of low quality or have a high risk of bias. The chosen articles were of moderate to high grade (Table 1).

2.8 Study Selection

Mendeley (Elsevier, London, UK) was used to remove duplicate publications. Two writers (KA and SA) independently examined the abstracts using Rayyan software. Any disagreements between the two writers' assessments of the abstracts that were generated were settled by a third author (MA). Complete texts of potentially pertinent studies were subsequently evaluated. The agreement rate between the two reviewers was evaluated using Cohen's kappa coefficient throughout all methodological phases. The rate of 0.83 is regarded as "strong agreement" [15, 19].

2.9 Data Analysis

To ascertain the prevalence, characteristics, treatment modality, and outcomes of diabetic foot ulcer in Nigeria, we carried out a narrative synthesis. Two independent reviewers evaluated the quality of the chosen papers using the Newcastle–Ottawa scale [20].

To estimate the overall pooled prevalence of diabetes foot ulcers, its distributions across the geopolitical zones, and the treatment outcomes, a random effect meta-analysis was performed based on the degree of heterogeneity. We then calculated the pooled diabetic foot ulceration prevalence and its 95% confidence intervals (CI). We utilize I^2 test statistics to calculate the heterogeneity effect and evaluate the consistency of investigations. Consequently, a random effect model was required due to the heterogeneity between the first investigations ($I^2 = 97.1\%$, $p < 0.001$) [21]. An indicator of study heterogeneity, the I^2 statistic has values of 25%, 50%, and 75%, respectively, and can be classified as

Table 1 Supplemental digital content Newcastle-Ottawa quality assessment form for studies

Author/Year	Newcastle-Ottawa Scale									Total	Quality
	Selection				Comparability		Outcome				
	1 ^a	2 ^b	3 ^c	4 ^d	5 ^e	6 ^f	7 ^g	8 ^h	9 ⁱ		
Oboiren	●	●	●	●	●	●	●	●	●	9	Good
Ogbera	●	●	●	●	●	●	●	-	●	8	Good
Isiguzo	●	●	●	●	●	●	●	-	●	8	Good
Aliyu	●	●	●	●	●	●	●	-	●	8	Good
Enweluzo	●	●	●	●	●	●	●	-	●	8	Good
Ezeani	●	●	●	●	●	●	●	-	●	8	Good
Habibu	●	●	●	●	●	●	●	-	●	8	Good
Danmusa	●	●	●	●	-	-	●	-	●	6	Fair
Salawu	●	●	●	●	●	●	●	●	●	9	Good
Adeleye	●	●	●	●	●	●	●	-	●	8	Good
Unachukwu	●	●	●	●	●	●	●	-	●	8	Good
Ugwu	●	●	●	●	●	●	●	-	●	8	Good
Ekere	●	●	●	●	●	●	●	-	●	8	Good
Ismail	●	●	●	●	●	●	●	-	●	8	Good
Adeleye	●	●	●	●	●	●	●	-	●	8	Good
Ekpebegh	●	●	●	●	-	-	●	-	●	6	Fair
Ogbera	●	●	●	●	●	●	●	●	●	9	Good
Ajayi	●	●	●	●	●	●	●	-	●	8	Good
Edo	●	●	●	●	●	●	●	-	●	8	Good

^a Representativeness of the exposed cohort^b Selection of the non-exposed cohort^c Ascertainment of exposure^d Demonstration that outcome of interest was not present at start of study^e Comparability of cohorts on the basis of the design or analysis (adjusted for age)^f Comparability of cohorts on the basis of the design or analysis (adjusted for any other factor)^g Assessment of outcome^h Was follow-up long enough for outcomes to occurⁱ Adequacy of follow-up of cohorts

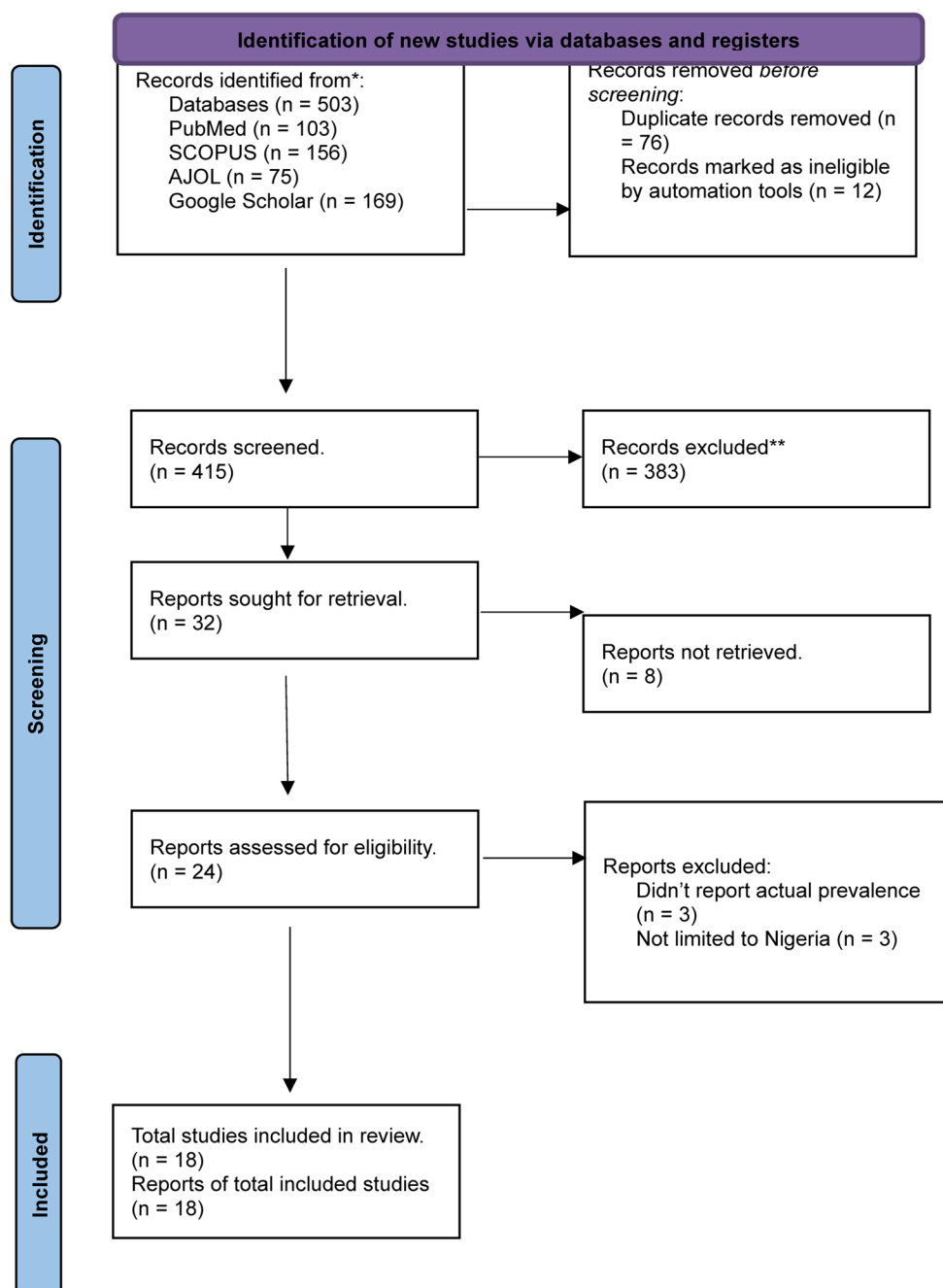
low, moderate, or high. The management results of diabetic foot ulcers were assessed using a forest plot and the effect size. Meta-regression and subgroup analysis were used to investigate the possible causes of heterogeneity. Deeks funnel plots and Begg's tests were employed to measure publication bias, as recommended by many researchers [22]. STATA (version 14.0, Stata Corporation, Texas, USA) was used for all statistical analyses, and a p-value of less than 0.05 was deemed statistically significant.

3 Result

Figure 1 shows the flow chart and selection process for the national prevalence of diabetic foot ulceration in Nigeria. A total of 18 articles were selected for the study (9–11, 31–45), including eleven cross-sectional studies (10, 32, 35–38, 39–41, 43, 44), three multicentre observational and retrospective studies each [9, 11, 34] and [22–24] respectively and one cohort study [25]. Studies

selected were between year 2005 to 2024, with a total of 2616 subjects with diabetic foot ulcers. Figure 2 represents the number of selected studies across the six geopolitical zones. A total of 13 studies reported the DFU by gender; 57.17% of subjects are males, representing a male-to-female ratio of 1.41:1. The mean age (+ SD) was 56.43 ± 2.99 years, and mean duration of diabetes among subjects was 10.62 ± 8.20 years, ranging from 0 to 30 years. Duration of diabetic foot ulcer before presentation ranged from 36 to 72 days, with a mean duration of 46.65 ± 14.65 days, as reported in five studies [6, 24–27]. Ten studies reported the type of diabetes in DFU, with type 2 diabetes mellitus (T2DM) being the most prevalent, 95.35% (95% CI 89.57–99.00.57.00) while 4.65% (95% CI 1.00–10.43) had type 1 diabetes mellitus (T1DM). Characteristics of the included studies are as summarized in supplementary Table 1. The quality assessment of included studies was done using the Newcastle–Ottawa Scale (NOS) and is as summarized in Table 2 and is included in the supplementary file.

Fig. 1 The flow chart and selection process for the global prevalence of diabetic foot ulceration



3.1 Pooled prevalence of diabetic foot ulcer

Eight studies reported the prevalence of DFU (1,3, 6–8,10,11,17). The pooled prevalence was 16.69% (95% CI. 9.74–25.06). There is a high degree of heterogeneity (I^2) of 97.43%. Subgroup analysis by geopolitical zone shows that DFU was most common in the Southwest, 20.38% (95% CI. 7.13–38.11) and lowest in the Northeast 5.98% (95% CI. 4.86–7.20). A Multicenter study reported the highest prevalence of 24.89% (95% CI. 22.62–27.23) as shown in Fig. 2.

3.2 Foot Affection

Among the three studies that reported the affected foot (33, 39,42), diabetic foot ulcer was to be more common in the left foot, 55.09% (95% CI. 35.12–74.28) with a low degree of heterogeneity (I^2) of 0.00%, followed by the right foot with a prevalence of 30.54% (95% CI. 11.95–52.97, I^2 =21.46%) and both feet 13.40% (95% CI. 0.65–35.87, I^2 =40.57%). Risk factors found to be associated with the development of diabetic foot ulcer are shown in Fig. 3. The most common risk factors for

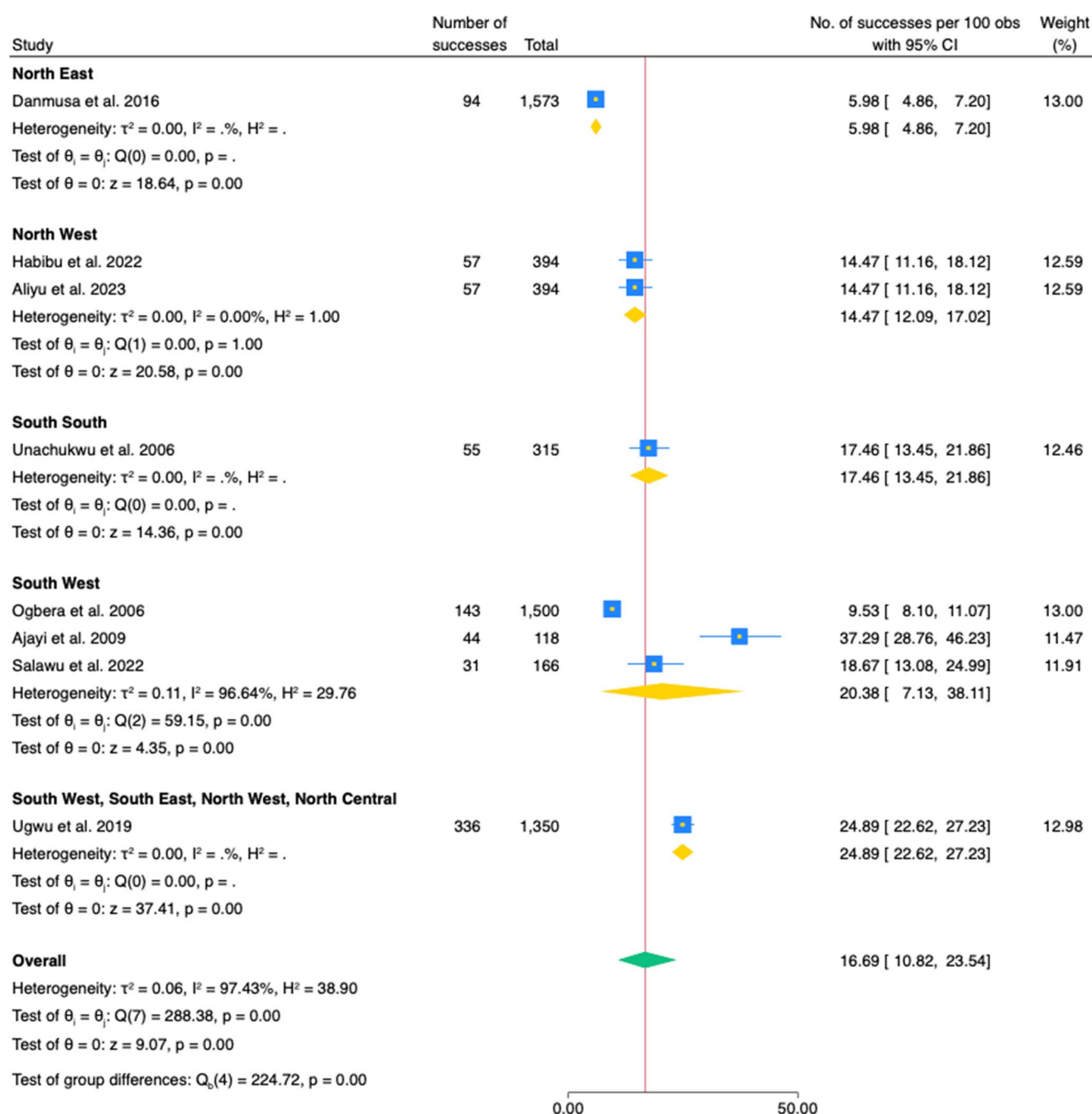


Fig. 2 Pooled prevalence of diabetic foot ulcer in Nigeria

DFU as reported were peripheral neuropathy, PAD, previous DFU, visual impairment and male gender among other factors.

3.3 Severity of Diabetic Foot Ulcer

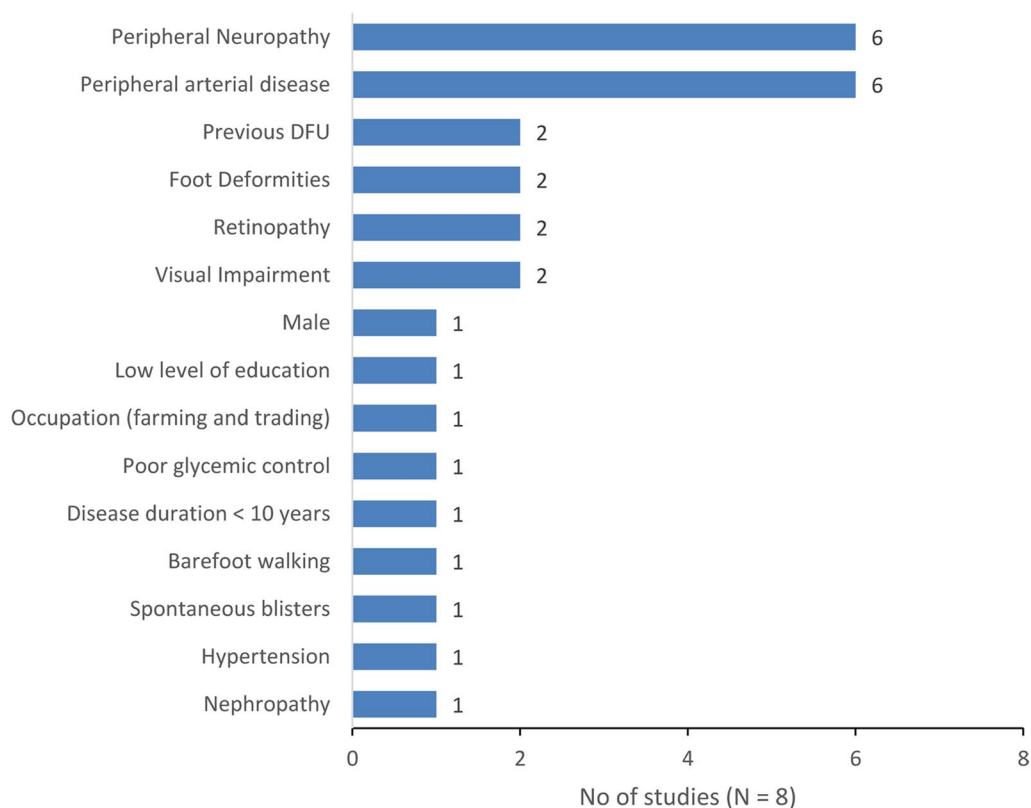
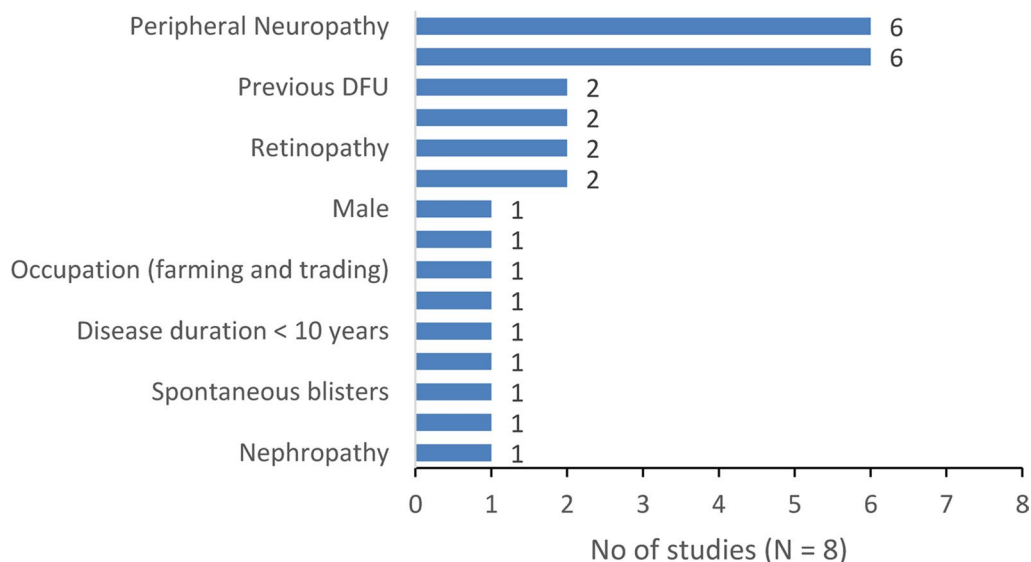
The severity of the diabetic foot ulcer was reported in twelve studies (9,11,33–35,38–40) using the Wagner staging method. Wagner stage IV was the most prevalent presentation among subjects, 26.67% (95% C.I.

17.52–36.90).and the occurrence of other classifications of DFU are as shown in Fig. 4.

3.4 Outcome of Diabetic Foot Ulcer

3.4.1 Amputation in DFU

Twelve studies reported amputation in patients with diabetic foot ulcer (9–11, 33, 34, 35, 39–43). The overall prevalence of amputation was 33.27% (95% CI. 23.09–44.29), there was

Fig. 3 Risk factors associated with diabetic foot ulcer**Fig. 4** Pooled prevalence of the diabetic foot ulcer based on Wagner staging

a high degree of heterogeneity ($I^2=96.22\%$). Subgroup analysis by geopolitical zone ranges from 19.44% (95% C.I. 7.90–34.19) in the Southeast to 56.86% (95% C.I. 42.99–70.23) in the Northwest (Fig. 5). The types of amputation done included above knee, below knee, trans-metatarsal, digital, Ray and disarticulation. Below knee amputation was the most common with a pooled prevalence of 39.81% (95% C.I. 8.21–76.20, $I^2=76.38\%$), then others (trans-metatarsal, digital, Ray and

disarticulation), 16.82% (95% C.I. 0.01–47.11, $I^2=68.81\%$) and above knee 7.06% (95% C.I. 0.00–38.00, $I^2=76.43\%$).

3.5 Mortality in Diabetic Foot Ulcer

Twelve studies reported the estimated mortality among the subjects (9–11,33,35,36,40,41,43,44,46,47). The pooled prevalence of mortality was 17.75% (95% CI

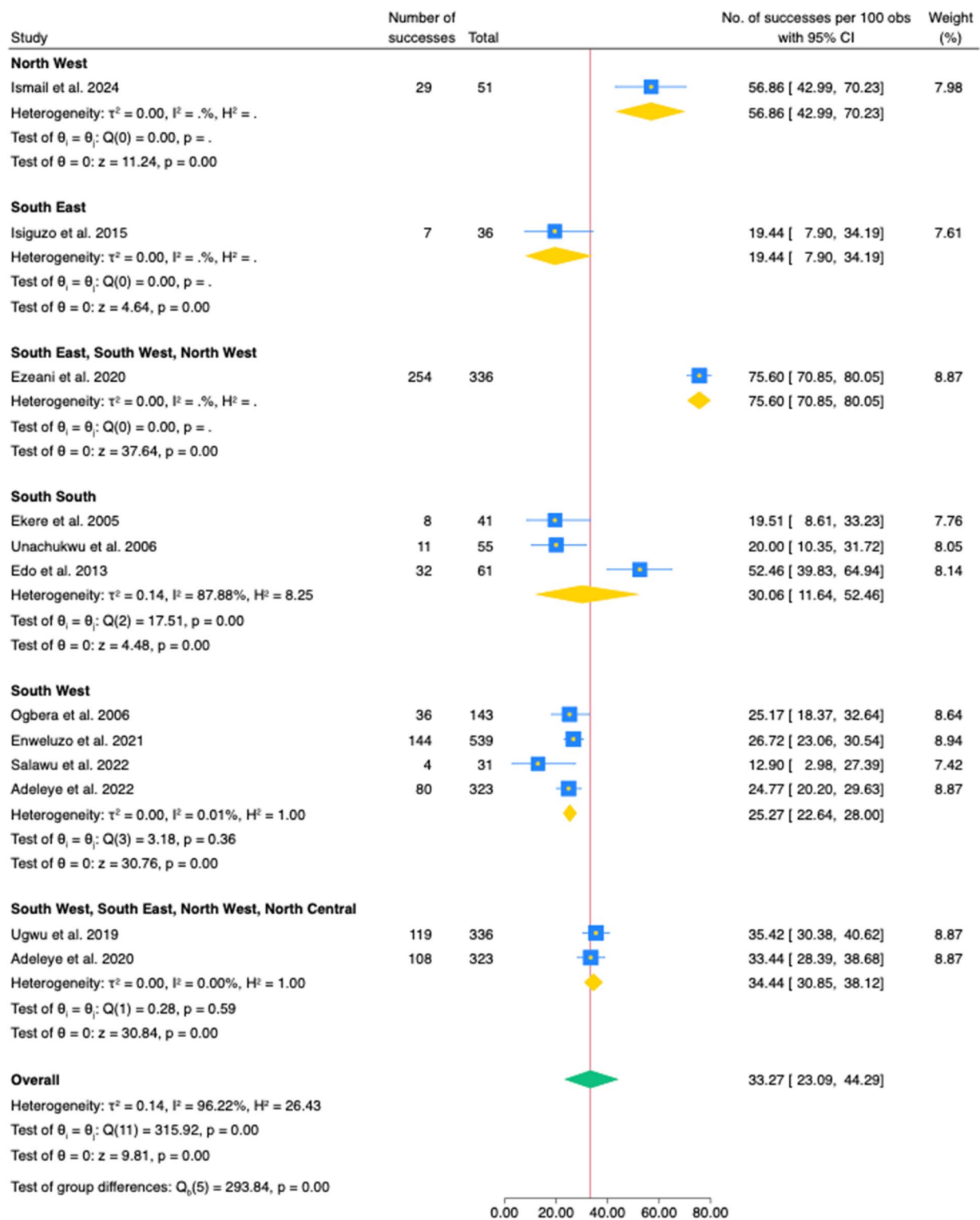


Fig. 5 pooled prevalence of Amputation among subjects with diabetic foot ulcer

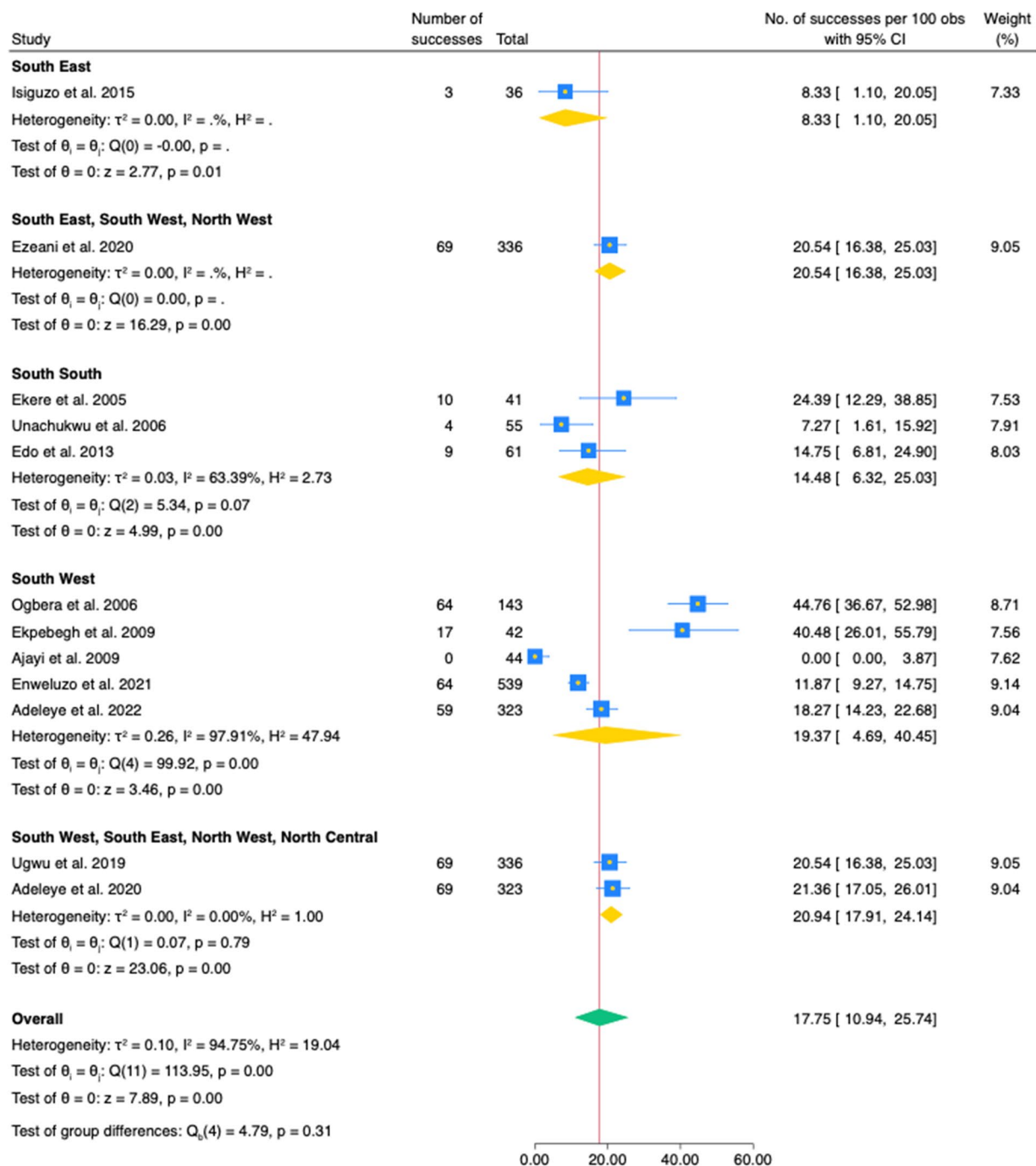


Fig. 6 Pooled prevalence of mortality among diabetic foot ulcer patients

10.94–25.74), with a high degree of heterogeneity (I^2 of 94.75%). Duration of hospital stay was reported in 5 studies, with a mean of 39.97 ± 10.83 days. The mortality rate ranges from 8.33% in the Southeast to 19.37% in the Southwest. The multicentre studies reported higher

mortalities of 20.54% and 20.94% (Fig. 5). A pooled prevalence of 43.32% (95% CI 28.71–58.53, $I^2 = 95.24\%$) subjects were discharged following treatment, while 11.52% (95% CI 4.77–20.51, $I^2 = 92.46\%$) subjects left hospital against medical advice (Fig. 6).

4 Discussion

Our systematic review included a moderate size of 2616 subjects from 18 studies in the past two decades. This study obtained pooled prevalence of diabetic foot ulcers in Nigeria to be 16.69% which varies from region to region. The pooled prevalence in this meta-analysis is slightly higher than the overall pooled prevalence of 13.0% in Africa [23], but much higher than the reported global prevalence of 6.3% [3]. However, this prevalence was lower than a study conducted in Khartoum Sudan, 18.1% [28] and Spain 17.4% [29]. This discrepancy could be due to the difference in the study method, or a reflection of variation in the demographic characteristics of the population investigated. It may also be related to the cultural practice of people walking barefoot especially those in the rural areas, or to the poor health seeking behaviour of patients, or poor disposable income as more than 90% of Nigerians patients pay out-of-pocket [30] due to low national health insurance coverage in Nigeria. The results showed that the highest prevalence of diabetic foot ulceration was reported in Southwest (16.69%), and the lowest prevalence was reported in the Northwest (5.98%). The regional variation in the prevalence of diabetic foot ulcer may be related to differences in cultural practices and the number of studies used in the analysis.

Our data suggested that diabetic foot ulcers and the incidences of minor and major amputation are higher among men. This is in keeping with a finding by Salawu et al. [31] who found that male were about five times more likely to have DFU than female patients. This is also consistent with findings from another study in Bangladesh [26]. One explanation for this gender difference might be that males are more likely to engage in outdoor activities or be involved in increased physical work that may result in trauma to the feet which may not heal if the patient has background diabetes. Sex differences are also likely be explained by underlying risk factors, access to care, screening, and adherence to treatment as women are more sensitive to health issues and more likely to seek healthcare services and women are more likely to perform recommended self-care and foot care compared to men [27, 32, 33]. Men with diabetes have a higher prevalence of peripheral neuropathy, peripheral arterial disease, and cardiovascular disease which together account for most observed sex differences in DFU risk [25]. The occurrence of DFU more on the left foot compared to the right might be due to more men were right-footed. Hence, right foot is likely to be less injured than the left.

We also found differences in amputation rates within different geopolitical regions of Nigeria. This meta-analysis found higher diabetes-related amputation rate in the Southern part of Nigeria compared to most regions of the Northern Territory. The possible reasons for this are not clearly

known but it could be due to poor health seeking behaviour of subjects in the south and higher number of orthopaedic surgeons and facilities in the southern part of the country. Poor glycaemic control, diabetes for a longer period of time, advanced age, peripheral neuropathy, peripheral vascular disease, diabetes kidney disease (DKD), and delayed hospital presentation after DFU development are among the numerous risk variables that are consistently linked to increased amputation rates. Due to inadequate perfusion and healing rates as well as the peripheral vascular disease process, ischaemic ulcer type was revealed to have a substantial impact on the total amputation and mortality rate among DFU patients. Also, infected wounds, among others contributed greatly to these undesirable results. Lavery et al. [27] showed that amputation was 154 times more likely in infected wounds and that the length of an ulcer over 30 days increased the likelihood of wound infection by almost five times. This is probably due to the higher propensity of accelerated tissue necrosis and gangrene in such wounds especially in a limb with compromised vascular supply. Ugwu et al. [34] reported that wound infection was present in 76.8% of studied patients while over half had developed some form of gangrene at presentation in the hospital. Such patients are expected to suffer more amputation and death from overwhelming sepsis.

In term of diabetes severity as assessed by Wagner staging, most patients (at least 7 in 10) with DFU presented with stage ≥ 3 , meaning, presented with advanced stage. The delayed presentation of the non-healing wound was attributed as the outcome of wrong belief of “stepping on poison” or “spiritual attack”, hence, orthodox care was usually not sought until the disease was advanced. This late hospital presentation was a common denominator in many previous studies of DFU in Nigeria [5, 6, 25, 35–37]. The ulcer had lasted more than 1 month in over 70% of the patients prior to hospitalization, hence presenting in advanced stage. This may partly be responsible for a relatively high percentage of amputations (approximately 1 in 5 patients) and increased mortality. A significant association between ulcer severity as measured by Wagner grading, and amputation as well as mortality were reported by many studies in Nigerian patients with DFU [25, 36, 37]. Patients with advanced Wagner grade ulcers ($\geq$ grade 3) were almost eight times more likely to die compared to those with lower grade ulcers. It should be noted that higher Wagner foot ulcer grade indicates a more advanced disease and increased probability of sepsis which could predispose the patient to multi-organ dysfunction. Importantly, appropriate multi-disciplinary care team approach led by an endocrinologist has been found to drastically improve diabetic foot outcomes and is hereby advocated [24].

Amputation and intra-hospital mortality rates of one in three patients (33.27%) and one in five patients (17.75%) respectively that were observed in this study are significantly higher than the incidence reported in developed countries. For instance, in Australia, lower limb amputation rate from DFU is less than 2% with over 70% being minor amputation [38]. Very low amputation and mortality rates were also reported in Netherlands, Thailand, Scotland, Manchester, England, and United States (US) [39–43]. Differences in the quality of healthcare systems and cultural practices are likely to be responsible for these discrepancies. Over the same time, major amputation made up an increasing proportion of these amputations, implying to lower rates of lower-extremity revascularization procedures in Nigeria; these trends are thought to be a challenge to improved efforts at limb preservation. This is contrary to findings in developed countries where minor amputations made up the major proportion of these amputations [3].

4.1 Strengths and Limitations

This meta-analysis has many important limitations. It included various study designs cross-sectional, retrospective, multicentre observational, and prospective—across various settings (population-, hospital-, community-based, and public health centres), contributing to significant heterogeneity (>90%). Although an extensive search strategy using four databases was applied, grey literature was not included, possibly omitting relevant government reports. However, grey literature is often not peer-reviewed and difficult to retrieve using reproducible methods. Hence, this may not have significantly affected the result. Additionally, many studies were retrospective, relying on pre-recorded data, and most were hospital-based located in urban areas, limiting generalisability to rural or community settings. The focus on high-risk patients referred to hospitals suggests that many individuals with diabetic foot ulcers (DFUs) may have been excluded.

Despite these limitations, this study has much significant strength. This study is the first to comprehensively determine the epidemiology and DFU-related outcomes and provides the best evidence to date on the magnitude of DFU in Nigeria. Second, this review followed established protocols, validated the search strategy, and involved independent reviewers using a standardised quality assessment tool. These methodological strengths enhance the credibility of the findings.

We recommend that future large-scale, multicentre studies using internationally validated DFU outcome definitions and include nationally representative populations be conducted. Such research is urgently needed

to better quantify the DFU burden in Nigeria. Current evidence indicates that Nigeria faces a significant DFU burden, with high rates of hospitalisation, amputation, and mortality compared to global estimates.

5 Conclusion

In conclusion, this study found a high prevalence of diabetic foot ulceration in Nigeria with high mortality. In addition, diabetic ulcers developed at a younger age than reported in developed countries. Moreover, rates of ulcer, amputation, and mortality are affected by late diabetes diagnosis and late presentation in the hospital probably due to patients' perverted belief as to the cause of foot ulcers and poverty. This study represents the first nationwide epidemiological data on diabetic foot ulceration. In addition, deliberate and improved efforts focusing on DFU prevention should be integrated to the routine management process of diabetic patients and important steps required to reduce the burden of this devastating complication of diabetes.

Abbreviations

BMI	Body mass index
DM	Diabetes mellitus
FPG	Fasting plasma glucose
HbA1c	Glycated haemoglobin
IDF	International Diabetes Federation
NCDs	Non-communicable diseases
OGTT	Oral glucose tolerance test
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
RCT	Randomized controlled trial
T2DM	Type 2 diabetes mellitus

Acknowledgements Not applicable.

Author Contributions MAO was responsible for the conceptualization and the design of the research. Screening of the titles with abstracts quantitative analysis of selected studies and data analysis were done by SAA and KA. MAO, SAA and KA wrote the initial draft of the manuscript. All authors including TA and OBO contributed to the discussion and gave their approval for the final version of the manuscript.

Funding No funding was received for this study.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Ethical Approval and Consent To Participate Not applicable.

Consent for Publication Not applicable.

Competing Interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

References

- International Diabetes Federation. IDF Diabetes Atlas, 11th edn. Brussels, Belgium. 2019. Available from: <https://www.diabetesatlas.org>. Accessed July 25, 2024.
- Armstrong DG, Boulton AJ, Bus SA. Diabetic foot ulcers and their recurrence. *N Engl J Med*. 2017;376(24):2367–75.
- Pickwell K, Siersma V, Kars M, Apelqvist J, Bakker K, Edmonds M, et al. Predictors of lower-extremity amputation in patients with an infected diabetic foot ulcer. *Diabetes Care*. 2015;38(5):852–7.
- Kerr M, Barron E, Chadwick P, Evans T, Kong WM, Rayman G, et al. The cost of diabetic foot ulcers and amputations to the National Health Service in England. *Diabet Med*. 2019;36(8):995–1002.
- Ogbera AO, Fasanmade O, Ohwovoriole AE, Adediran O. An assessment of the disease burden of foot ulcers in patients with diabetes mellitus attending a teaching hospital in Lagos, Nigeria. *Int J Low Extrem Wounds*. 2006;5:244–9.
- Skrepnek GH, Mills JL Sr, Armstrong DG. A diabetic emergency one million feet long: disparities and burdens of illness among diabetic foot ulcer cases within emergency departments in the United States, 2006–2010. *PLoS One*. 2015;10(8):e0134914. <https://doi.org/10.1371/journal.pone.0134914>.
- Rice JB, Desai U, Cummings AKG, Birnbaum HG, Skornicki M, Parsons NB. Burden of diabetic foot ulcers for medicare and private insurers. *Diabetes Care*. 2014;37(3):651–8.
- Walsh JW, Hoffstad OJ, Sullivan MO, Margolis DJ. Association of diabetic foot ulcer and death in a population-based cohort from the United Kingdom. *Diabet Med*. 2016;33(11):1493–8.
- Abbas ZG, Boulton AJ. Diabetic foot ulcer disease in African continent: 'from clinical care to implementation' – review of diabetic foot in last 60 years – 1960 to 2020. *Diabetes Res Clin Pract*. 2022;183:109155.
- Goodridge D, Trepman E, Embil JM. Quality of life of adults with unhealed and healed diabetic foot ulcers. *Foot Ankle Int*. 2006;27(4):274–80.
- Beeson SA, Neubauer D, Calvo R, Sise M, Martin M, Kauvar DS, et al. Analysis of 5-year mortality following lower extremity amputation due to vascular disease. *Plast Reconstr Surg*. 2023;111(1):e4727. <https://doi.org/10.1097/GOX.00000000000004727>.
- Udo KR, Ajayi JFA, Kirk-Greene AHM, Falola TO. Nigeria. *Encyclopedia Britannica* [Internet]. 2020. Available from: <https://www.britannica.com/place/Nigeria>. Accessed July 5, 2021.
- Worldometer. Nigeria Population (2020) – Worldometer [Internet]. www.worldometers.info/2023. Available from: <https://www.worldometers.info/world-population/nigeria>
- population/#:~:text=The%20current%20population%20of%20Nigeria .
- United Nations Population Fund. UNFPA - United Nations Population Fund [Internet]. [unfpa.org](https://www.unfpa.org/). 2021. Available from: <https://www.unfpa.org/>
- Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*. 2021;372:n71. <https://doi.org/10.1136/bmj.n71>.
- Munn Z, Stern C, Aromataris E, Lockwood C, Jordan Z. What kind of systematic review should I conduct? A proposed typology and guidance for systematic reviewers in the medical and health sciences. *BMC Med Res Methodol*. 2018;18(1):5.
- Moola S, Munn Z, Tufanaru C, Aromataris E, Sears K, Sfetcu R, et al. Chapter 7: systematic reviews of etiology and risk. In: Aromataris E, Munn Z, editors. *Joanna Briggs Institute Reviewer's Manual*. Volume 5. The Joanna Briggs Institute; 2017. p. 219–71.
- McHugh ML. Interrater reliability: the kappa statistic. *Biochem Med (Zagreb)*. 2012;22(3):276–82.
- Gierisch JM, Beadles C, Shapiro A et al. Health disparities in quality indicators of healthcare among adults with mental illness [Internet]. Washington (DC): Department of Veterans Affairs (US); 2014 Oct. Appendix B, Newcastle-Ottawa Scale Coding. Manual for Cohort Studies. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK299087/>
- Higgins JP, Thompson SG, Deeks JJ, Altman DG. Measuring inconsistency in meta-analyses. *BMJ*. 2003;327(7414):557–60.
- Dear KB, Begg CB. An approach for assessing publication bias prior to performing a meta-analysis. *Stat Sci*. 1992 May;1:237–45.
- Rigato M, Pizzol D, Tiago A, Putoto G, Avogaro A, Fadini GP. Characteristics, prevalence, and outcomes of diabetic foot ulcers in Africa: a systematic review and meta-analysis. *Diabetes Res Clin Pract*. 2018;142:63–73.
- Dòria M, Rosado V, Pacheco LR, Hernández M, Betriu À, Valls J, et al. Prevalence of diabetic foot disease in patients with diabetes mellitus under renal replacement therapy in Lleida, Spain. *Biomed Res Int*. 2016;2016:7217586.
- Rubio JA, Aragón-Sánchez J, Jiménez S, Guadalix G, Albarracín A, Salido C, et al. Reducing major lower extremity amputations after the introduction of a multidisciplinary team for the diabetic foot. *Int J Low Extrem Wounds*. 2014;13(1):22–6.
- Ekpebegh CO, Iwuala SO, Fasanmade OA, Ogbera AO, Igumbor E, Ohwovoriole AE. Diabetes foot ulceration in a Nigerian hospital: in-hospital mortality in relation to the presenting demographic, clinical and laboratory features. *Int Wound J*. 2009;6(5):381–5.
- Mansur CL, Rustveld LO, Nash SG, Jibaja-Weiss ML. Social factors and barriers to self-care adherence in Hispanic men and women with diabetes. *Patient Educ Couns*. 2015;98(6):805–10.
- Lavery LA, Armstrong DG, Wunderlich RP, Mohler MJ, Wendel CS, Lipsky BA. Risk factors for foot infections in individuals with diabetes. *Diabetes Care*. 2006;29(6):1288–93.
- Almobarak AO, Awadalla H, Osman M, Ahmed MH. Prevalence of diabetic foot ulceration and associated risk factors: an old and still major public health problem in Khartoum, Sudan? *Ann Transl Med*. 2017;5(17):1–5.
- Salawu AI, Ajani GO, Soje MO, Ojo OD, Olatinri EO, Obajolowo OO, et al. Diabetes mellitus foot ulcer and associated factors among type 2 diabetes patients in a tertiary institution in Southwest Nigeria. *Ann Afr Med*. 2022;21(4):339–47.
- Banik PC, Barua L, Moniruzzaman M, Mondal R, Zaman F, Ali L. Risk of diabetic foot ulcer and its associated factors among Bangladeshi subjects: a multicentric cross-sectional study. *BMJ Open*. 2020;10(2):e034058. <https://doi.org/10.1136/bmjopen-2019-034058>.
- Rossaneis MA, Haddad MD, Mathias TA, Marcon SS. Differences in foot self-care and lifestyle between men and

- women with diabetes mellitus. *Rev Latino-Am Enfermagem*. 2016;24:e276124.
32. Tang ZQ, Chen HL, Zhao FF. Gender differences of lower extremity amputation risk in patients with diabetic foot: a meta-analysis. *Int J Low Extrem Wounds*. 2014;13(3):197–204.
33. Sorber R, Abularrage CJ. Diabetic foot ulcers: epidemiology and the role of multidisciplinary care teams. *Semin Vasc Surg*. 2021;34:47–53.
34. Ugwu E, Adeleye O, Gezawa I, Okpe I, Enamino M, Ezeani I. Burden of diabetic foot ulcer in Nigeria: current evidence from the multicenter evaluation of diabetic foot ulcer in Nigeria. *World J Diabetes*. 2019;10(3):200–11.
35. Adeleye OO, Ugwu ET, Gezawa ID, Okpe I, Ezeani I, Enamino M. Predictors of intra-hospital mortality in patients with diabetic foot ulcers in Nigeria: data from the MEDFUN study. *BMC Endocr Disord*. 2020;20(1):1–10.
36. Anumah FO, Mshelia-Reng R, Abubakar A, Sough T, Asudo F, Jamda MA, et al. Management outcome of diabetic foot ulcers in a teaching hospital in Abuja. *Nigeria Age (Years)*. 2017;54(1304):21–90.
37. Edo AE, Edo GO, Ezeani IU. Risk factors, ulcer grade and management outcome of diabetic foot ulcers in a tropical tertiary care hospital. *Niger Med J*. 2013;54(1):1–5.
38. Baba M, Davis WA, Norman PE, Davis TM. Temporal changes in the prevalence and associates of diabetes-related lower extremity amputations in patients with type 2 diabetes: the Fremantle diabetes study. *Cardiovasc Diabetol*. 2015;14:1–10.
39. Senneville É, Albalawi Z, van Asten SA et al. Guidelines on the diagnosis and treatment of foot infection in persons with diabetes: IWGDF/IDSA 2023. Published 2023. Available from: <https://iwgdfguidelines.org/wp-content/uploads/2023/05/IWGDF-2023-04-Infection-Guideline.pdf>. Accessed June 12, 2023.
40. Stoekenbroek RM, Lokin JL, Nielen MM, Stoes ES, Koelmay MJ. How common are foot problems among individuals with diabetes? Diabetic foot ulcers in the Dutch population. *Diabetologia*. 2017;60:1271–5.
41. Thewjitcharoen Y, Krittiyawong S, Porramatikul S, Parksook W, Chatapat L, Watchareejirachot O, et al. Outcomes of hospitalized diabetic foot patients in a multidisciplinary team setting: Thailand's experience. *Journal of Clinical & Translational Endocrinology*. 2014;1(4):187–91.
42. Kennon B, Leese GP, Cochrane L, Colhoun H, Wild S, Stang D. Reduced incidence of lower-extremity amputations in people with diabetes in Scotland: a nationwide study. *Diabetes Care*. 2012;35(12):2588–90.
43. Oyibo SO, Jude EB, Tarawneh I, Nguyen HC, Armstrong DG, Harkless LB. The effects of ulcer size and site, patient's age, sex and type and duration of diabetes on the outcome of diabetic foot ulcers. *Diabet Med*. 2001;18(2):133–8.
44. Tolossa T, Mengist B, Mulisa D, Fetensa G, Turi E, Abajobir A. Prevalence and associated factors of foot ulcer among diabetic patients in Ethiopia: a systematic review and meta-analysis. *BMC Public Health*. 2020;20:1–14.
45. Salad AM, Ibrahim N, Mohamed AE, Shafie ZM, Mohamud AH, Abdirashid A, et al. Prevalence of diabetic foot ulcers and associated factors among adult diabetic patients in three referral hospitals in Mogadishu, Somalia. *Front Public Health*. 2023;11:1195483.
46. Petersen BJ, Linde-Zwirble WT, Tan TW, et al. Higher rates of all-cause mortality and resource utilization during episodes-of-care for diabetic foot ulceration. *Diabetes Res Clin Pract*. 2022;184:109182.
47. Zhang P, Lu J, Jing Y, Tang S, Zhu D, Bi Y. Global epidemiology of diabetic foot ulceration: a systematic review and meta-analysis. *Ann Med*. 2017;49(2):106–16.
48. Behrendt CA, Sigvant B, Szeberin Z, et al. International variations in amputation practice: a VASCUNET report. *Eur J Vasc Endovasc Surg*. 2018;56(3):391–9.
49. Ramsey SD, Newton K, Blough D, McCulloch DK, Sandhu N, Reiber GE, et al. Incidence, outcomes, and cost of foot ulcers in patients with diabetes. *Diabetes Care*. 1999;22(3):382–7.
50. Fadare J, Olamoyegun M, Gbadegesin BA. Medication adherence and direct treatment cost among diabetes patients attending a tertiary healthcare facility in Ogbomosho, Nigeria. *Malawi Med J*. 2015;27(2):65–70.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.